



CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

ENVIRONMENTAL SCIENCE

AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions, TWO from each module.
2. Answer ALL questions.
3. Write your answers in the spaces provided in this booklet.
4. Do NOT write in the margins.
5. You may use a silent, non-programmable, scientific calculator to answer questions.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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MODULE 1

Answer BOTH questions.

1. (a) Ms Farmer wishes to cultivate a vegetable crop using a sustainable method. She was advised to consider using hydroponics.

- (i) Describe the practice of hydroponics.

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[2 marks]

- (ii) Explain FOUR benefits of hydroponics.

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[4 marks]

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- (b) Figure 1 shows the percentage contribution of agriculture by various roles to the economies of two Caribbean countries, A and B. Study Figure 1 and answer the questions that follow.

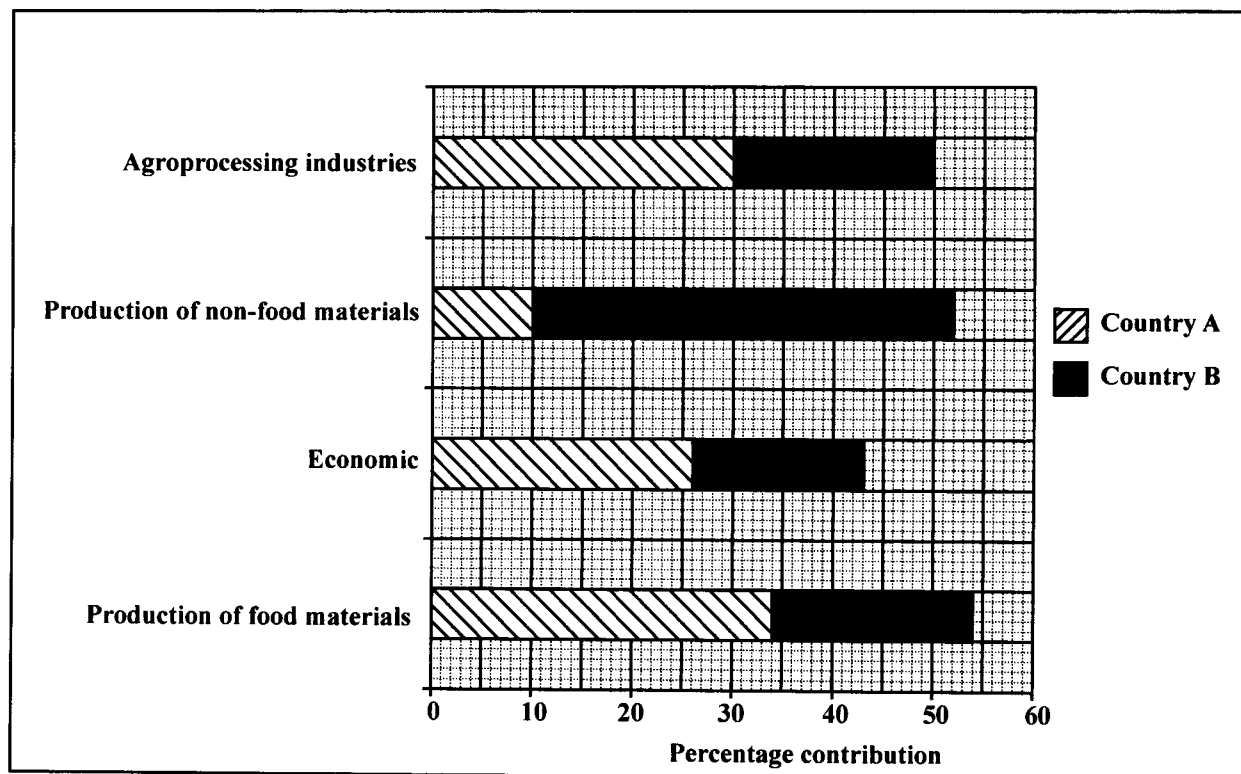


Figure 1. Percentage contributions made by the role of agriculture to the economies of two Caribbean countries

- (i) Make FOUR deductions about the contributions made by agriculture to the two countries.

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[4 marks]

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- (ii) In which country is the contribution of food security to the economy GREATER? Justify your answer.

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[4 marks]

- (c) Suggest THREE reasons why climate change is a major threat to sustainable agriculture in the Caribbean.

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Total 20 marks

2. (a) Figure 2 shows the percentage impact of the negative effects of agriculture in a country for two years, Year 1 and Year 2.

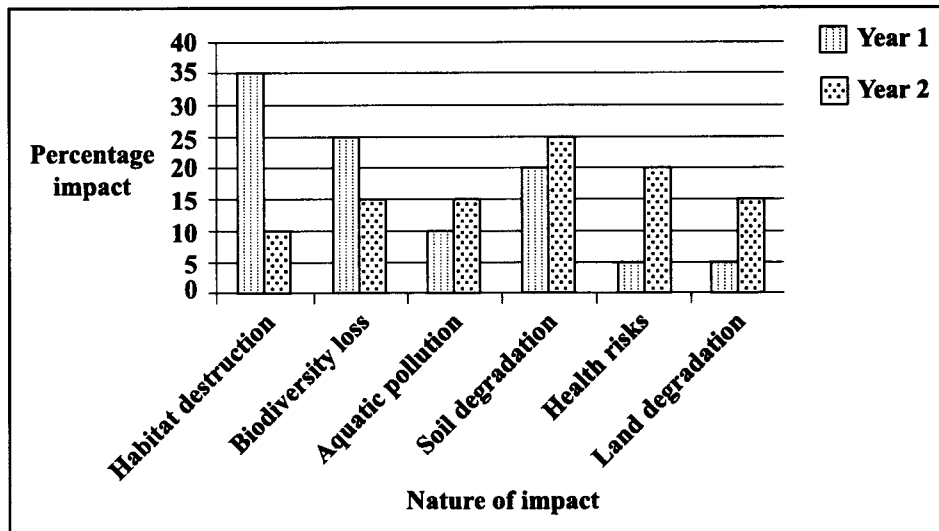


Figure 2. Impacts due to agriculture over a period of two years

- (i) Use Figure 2 to make FIVE deductions about the impact of the negative effects of agriculture in the country. Include specific values from Figure 2 in your responses.

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[5 marks]

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- (ii) Choose any THREE negative effects from Figure 2 and for EACH negative effect chosen, suggest ONE reason that can explain the difference in the percentage impact between Year 1 and Year 2.

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[6 marks]

- (b) Outline THREE characteristics of commercial agricultural systems.

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- (c) Using TWO examples, discuss how technology can be used to improve agricultural productivity.

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[6 marks]

Total 20 marks



MODULE 2

Answer BOTH questions.

3. (a) Outline the energy conversions that occur when a kettle boiling on a gas stove starts to whistle.

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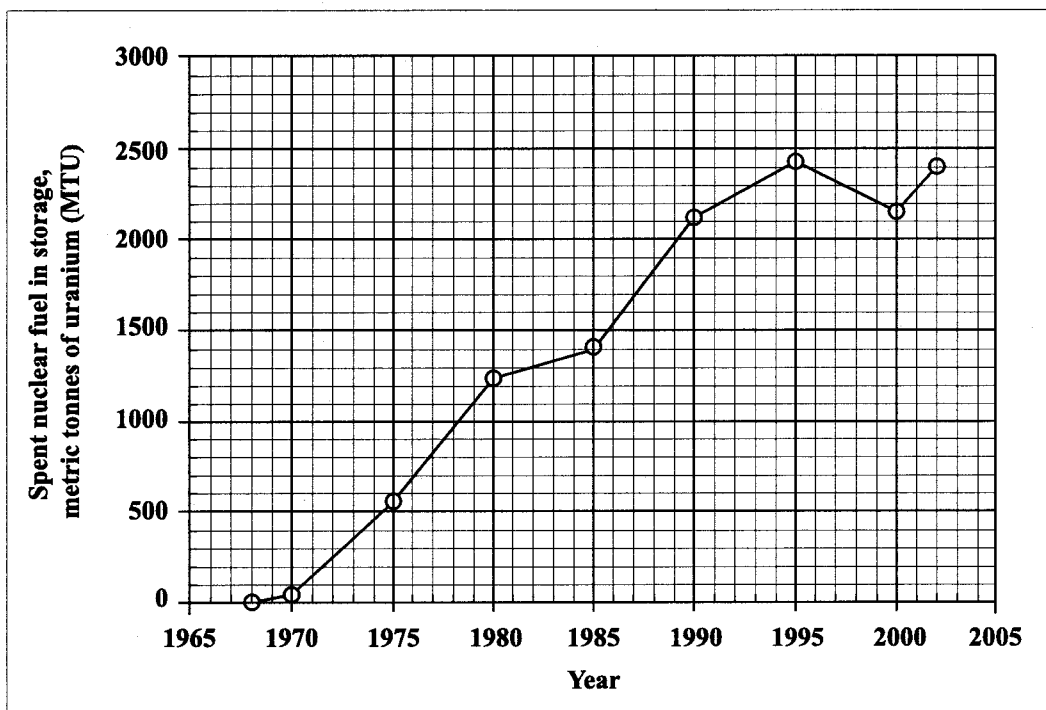
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[3 marks]

Figure 3 shows the amount of spent nuclear fuel in storage in the United States from 1968 to 2002. Carefully study Figure 3 and answer the questions that follow.



Source: The US Energy Information Administration, www.eia.gov.

Figure 3. Graph showing the amount of spent nuclear fuel in storage in the United States from 1968 to 2002

- (b) (i) Identify the MAJOR trend illustrated in Figure 3.

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[2 marks]

- (ii) How much spent nuclear fuel accumulated in the United States between 1985 and 2000?

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[3 marks]

- (c) Compare and contrast the generation of electricity by conventional and nuclear methods. In your answer provide ONE similarity and TWO differences.

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- (d) Outline THREE impacts of conventional electricity generation in the Caribbean.

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[6 marks]

Total 20 marks

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4. (a) Using ONE example, briefly explain the importance of energy to Caribbean societies.

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[3 marks]

- (b) (i) Define the term 'energy conservation'.

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- (ii) A Caribbean country would like to improve energy conservation in its transportation sector. Suggest ONE approach, explain how it would work to improve energy conservation and assess its suitability for the Caribbean.

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- (i) State the aim of this experiment.

[1 mark]

- [illegible]

[4 marks]

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- (d) A Caribbean country is considering switching from electricity generation using fossil fuel to hydropower. Evaluate the feasibility of this approach.

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[6 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions.

5. (a) (i) What is an environmental receptor?

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[2 marks]

- (ii) Suggest ONE example of an environmental receptor for noise pollution.

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[1 mark]

- (iii) A restaurant in your neighbourhood has nightly karaoke events that generate a lot of noise. Outline a method to measure the noise intensity at the restaurant. Clearly indicate how you would use this information to determine if the restaurant is in violation of noise pollution laws.

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- (b) Figure 4 shows a number of components in an ecosystem that are impacted by pesticides. Study Figure 4 and answer the questions that follow.

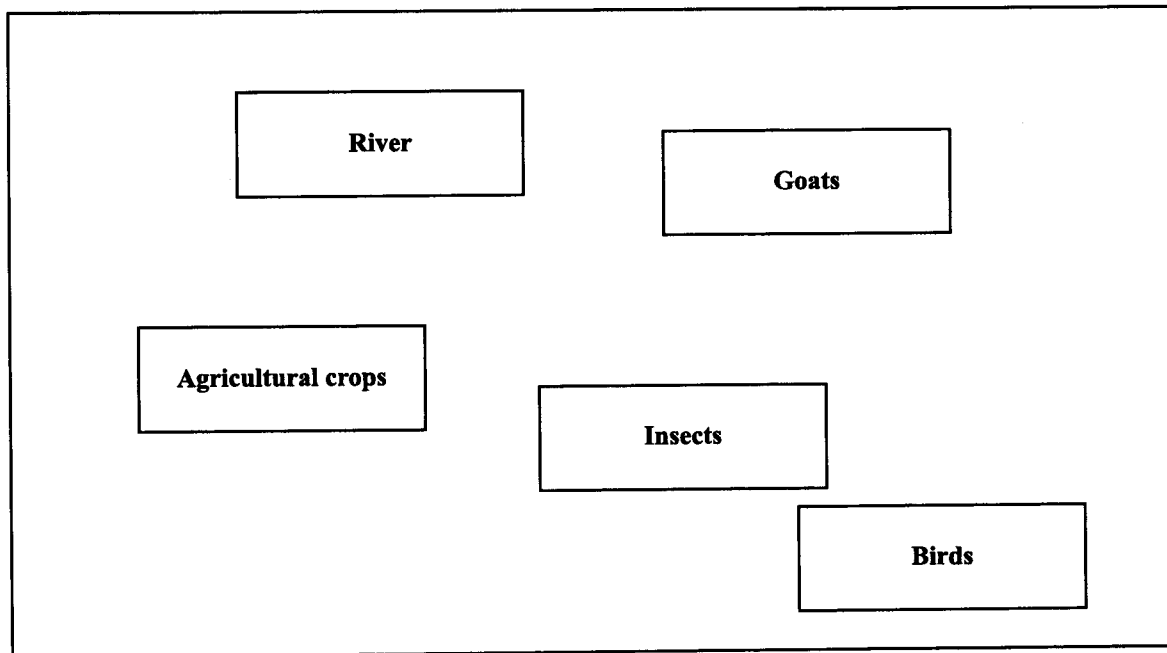


Figure 4. Diagram showing a selection of components in an ecosystem

- (i) On Figure 4, use arrows to indicate the likely movement of pesticides through the ecosystem. **[4 marks]**

- (ii) Which component is the starting point for the pollution?

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[1 mark]

- (iii) In which component would you expect to find the highest concentration of pesticides? Give a reason for your answer.

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- (c) The residents of a town are concerned that pollution may be impacting their drinking water. Explain how public awareness/education programmes AND technological processes can be used to mitigate the impacts of pollution on drinking water.

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6. (a) List THREE underlying causes of pollution.

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- (b) Table 1 shows the results of a programme to monitor atmospheric sulphur dioxide (SO_2) concentration in a Caribbean country over a one-year period. The calibration curve shown in Figure 5 can be used to convert colour intensity (absorbance) into SO_2 concentration (ppb). Study Figure 5 and Table 1 carefully and answer the questions that follow.

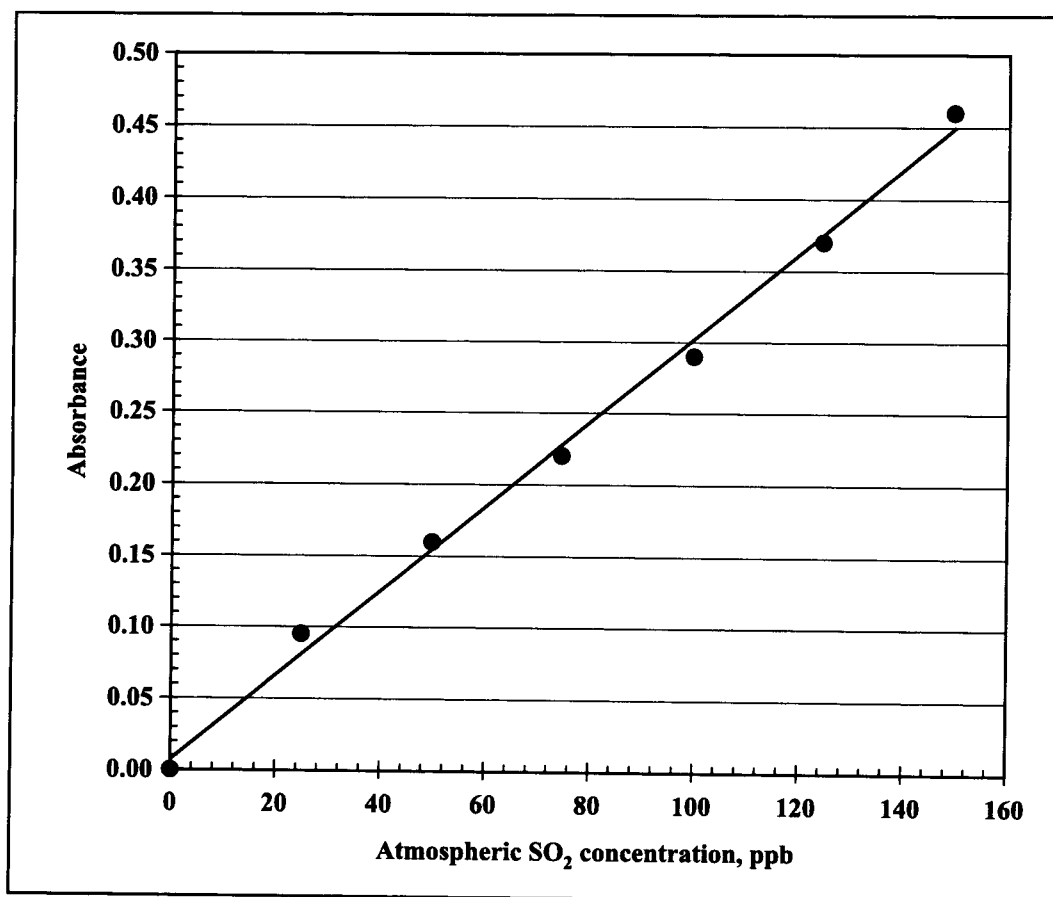


Figure 5. Calibration curve for determination of atmospheric SO_2 concentrations in a Caribbean country

TABLE 1: RESULTS OF A SO_2 MONITORING PROGRAMME IN A CARIBBEAN COUNTRY OVER A ONE-YEAR PERIOD

Date	Absorbance	Concentration of SO_2 (ppb)
January	0.36	
June	0.30	
December	0.23	

- (i) Complete Table 1 using the values in the calibration curve in Figure 5.



[3 marks]

- (ii) The environmental standards of the country indicate that an appropriate level of atmospheric SO_2 is 75 ppb. What conclusion can you make about the air quality in January with respect to SO_2 ? Give a reason for your response.

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[2 marks]

- (iii) In May, the Caribbean country implemented new legislation to reduce SO_2 emissions from automobiles. Evaluate the effectiveness of the legislation with the aid of the values in Table 1.

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[6 marks]

- (c) Why are international agreements required to solve air pollution problems like global warming and ozone depletion?

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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